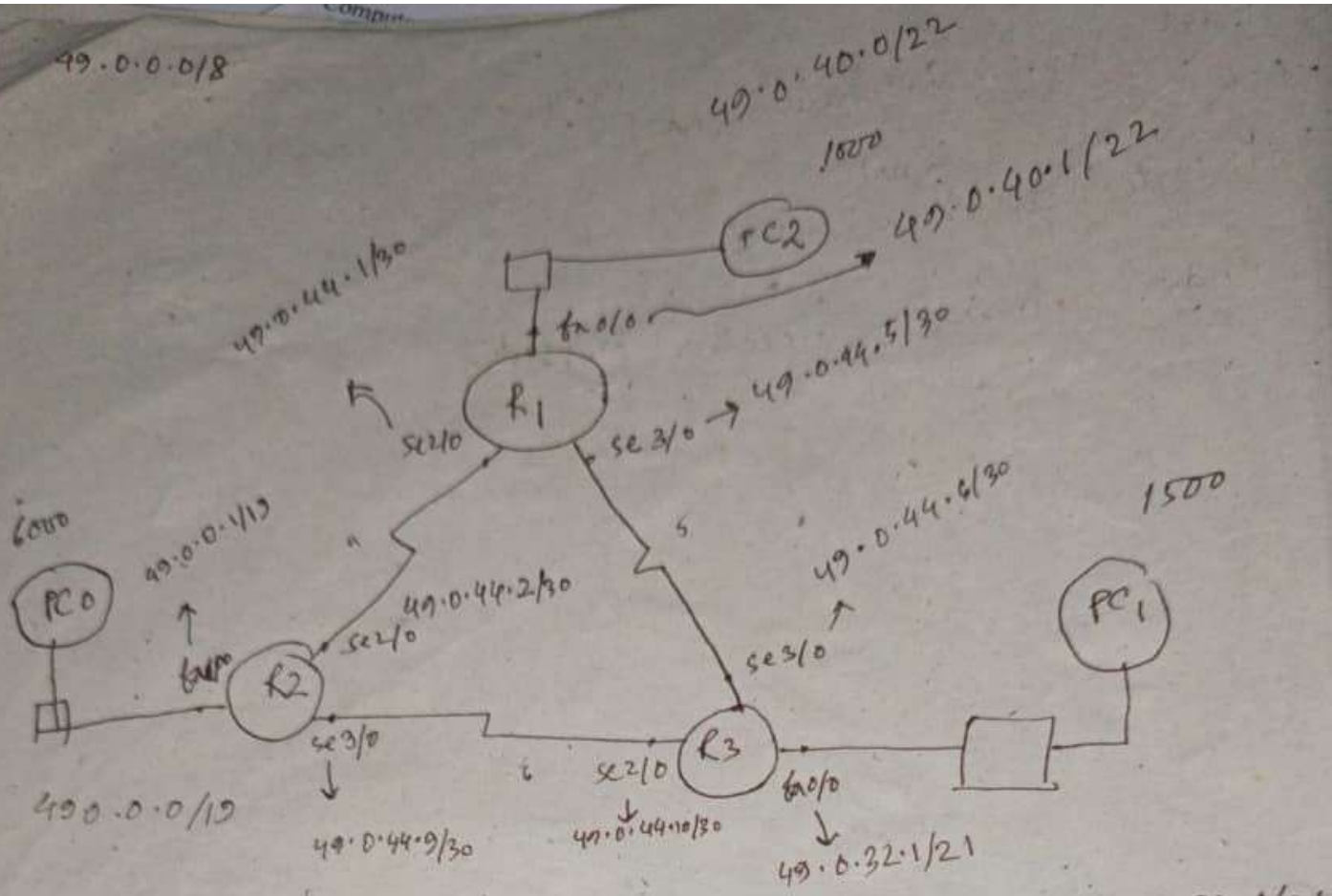


Subnet 1 $6000 \simeq 8192 = 2^{13}$ prefix portion = $32-13 = 19$

Subnet Mask = 255.255.224.0

Subnet Add = 49.0.0.0/19

Broadcast Add = 49.0.00011111.11111111

49.0.31.255/19

Subnet 2 $1500 \simeq 2048 = 2^{11}$ prefix portion = $32-11 = 21$

Subnet Mask = 255.255.248.0

Subnet Add = 49.0.32.0/21

Broadcast Add = 49.0.00100000.0 = 49.0.39.255/21

00100111 255

39

Subnet 3 1000 ≈ 1024 = 2^{10} prefix portion = 32 - 10 = 22

Subnet Mask = 255.255.252.0

Subnet Add = 49.0.40.0/22

Broadcast Add = 49.0.0010 1000.0
0010 1011 .255

49.0.43.255/22

Subnet 4: 2 Host ≈ 4 = 2^2 prefix portion = 32 - 2 = 30

Subnet Mask = 255.255.255.252

Subnet Add = 49.0.44.0/30

Broadcast Add = 49.0.44.3/30

Subnet 5: 2 Host ≈ 4 = 2^2 prefix portion = 32 - 2 = 30

Subnet Mask = 255.255.255.252

Subnet Add = 49.0.44.4/30

Broadcast Add = 49.0.44.7/30

Subnet 6: 2 Host ≈ 4 = 2^2 prefix portion = 32 - 2 = 30

Subnet Add = 49.0.44.8/30

Subnet Mask = 255.255.255.252

Broadcast Add = 49.0.44.11/30

Port Addressing

10.0.0.1
10.0.0.2
10.0.0.3
10.0.0.4

130.10.0.0

Private IP	Public IP
10.0.0.1/2001	130.10.0.0/4001
10.0.0.2/2002	130.10.0.0/4002
10.0.0.3/2003	130.10.0.0/4003
10.0.0.4/2004	130.10.0.0/4004

portno. range maximum: 65536

0-1024 fixed for server